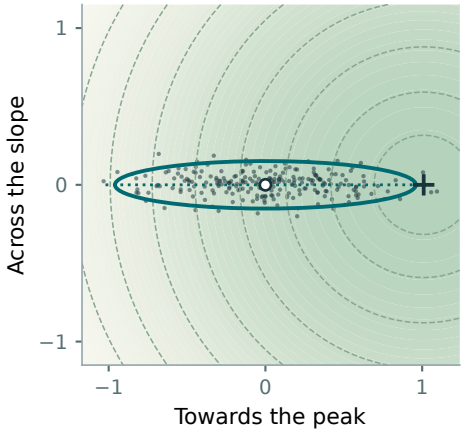
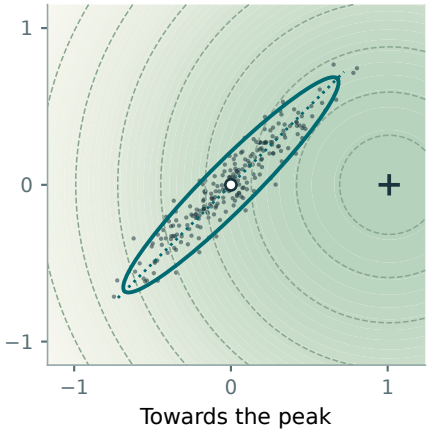
What changes when the breeding-value cloud turns?

The surface, population mean and two principal variances stay fixed. Only the direction of genetic variation changes.

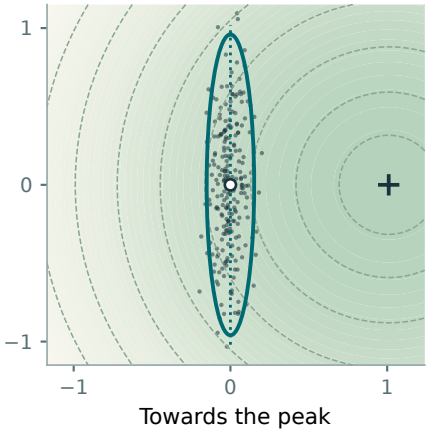
0° Towards the peak



45° Oblique to the slope

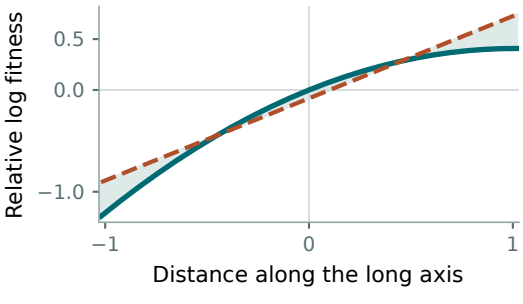


90° Across the slope

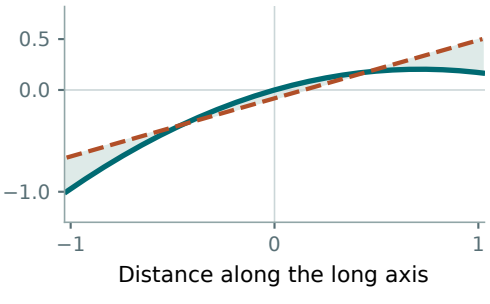


Side view along the long axis

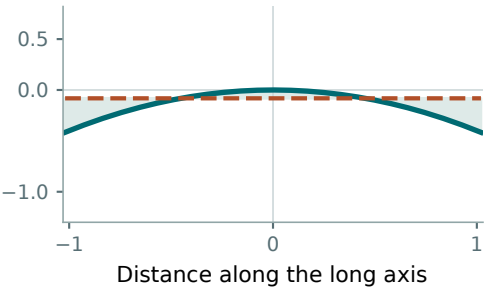
Solid: fitness surface · Dashed: fitted plane



$\mathcal{A}_g$ = **0.911**



$\mathcal{A}_g$ = **0.839**



$\mathcal{A}_g$ = **0.203**

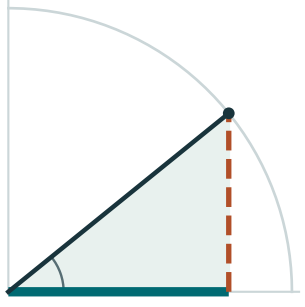
Dots are equally weighted parents. The ellipse contains 90% of the model population; + marks the fitness peak.

The index uses both traits. At 90° the slice is flat to first order, but the cloud still has some width towards the peak.

What does the angle measure?

Each triangle divides variation in log fitness into the part described by a plane and the part left by curvature.

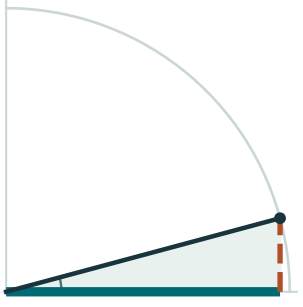
A Broad, on slope



Angle 39.0°

$\mathcal{A}_g$ = 0.604

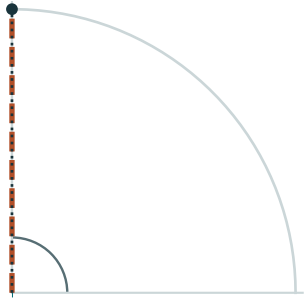
B Narrow, on slope



Angle 15.0°

$\mathcal{A}_g$ = 0.933

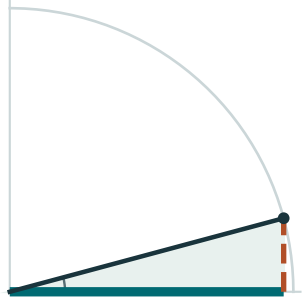
C Narrow, at peak



Angle 90.0°

$\mathcal{A}_g$ = 0.000

D Broad, farther away



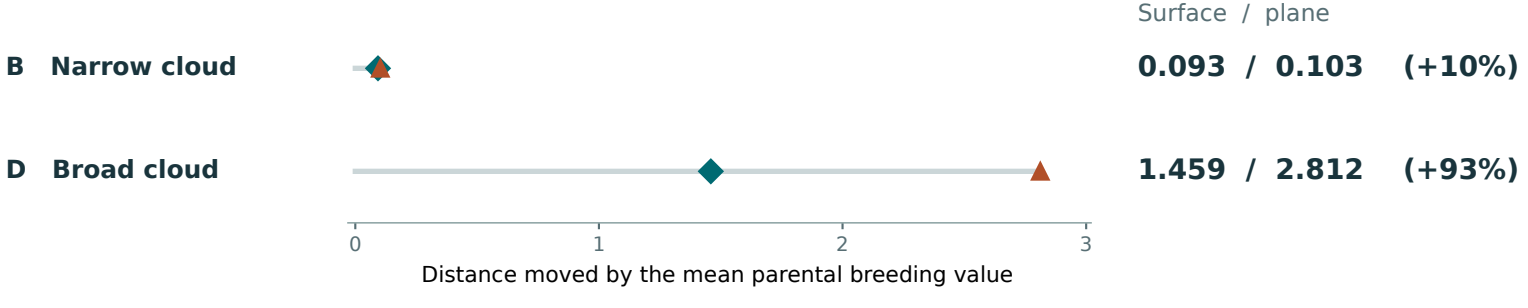
Angle 15.0°

$\mathcal{A}_g$ = 0.933

Base: linear SD | Height: curvature SD | Diagonal: total SD, scaled to 1 in every panel.

The squared base length gives the index: $\mathcal{A}_g = \cos^2(\psi)$. Here ψ is a projection angle, not the tilt of the fitness surface.

Now weight the same parents by their expected offspring



B and D have the same triangle. Their offspring-weighted means move different distances, in the same trait units.